

# NASRUL HUDA

MSc Data Science and Artificial Intelligence

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## PROFESSIONAL SUMMARY

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AI Engineer and Data Scientist with industry experience developing agentic AI workflows, RESTful backend services, and machine learning pipelines. Currently completing an MSc in Data Science and AI at the University of Hamburg. Track record of contributing to fast-moving teams and building solutions that are robust, well-tested, and ready to scale.

## EDUCATION

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### University of Hamburg

*Master of Science in Data Science and Artificial Intelligence*

### Jamia Millia Islamia

*Bachelor of Technology in Computer Science Engineering*

Hamburg, Germany

*Oct 2024 - Present*

New Delhi, India

*Oct 2020 - Jun 2024*

## EXPERIENCE

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### SynTwin GmbH

*AI Engineer*

Munich, Germany

*Oct 2025 - Mar 2026, Part Time*

- Developed agentic AI workflows and orchestration pipelines for a Digital Twin platform using Python and LiveKit.
- Implemented prompt engineering strategies to improve agent reliability across complex, multi-turn interactions.
- Built and maintained Vite React frontend components integrated with Supabase for real-time data sync and user authentication.
- Collaborated on end-to-end system design spanning AI agent architecture, backend services, and user-facing interfaces.
- **Technical Skills:** Python, TypeScript, Livekit, Supabase, Docker, Git, Github Actions

### University of Hamburg - Security and Distributed Systems Research Group

*Student Research Assistant*

Hamburg, Germany

*July 2025 - Sept 2025, Part Time*

- Built Python-based microservices for signing and verifying hash signatures as part of research data management and security analysis tooling.
- Contributed to the NFDIxCS (National Research Data Infrastructure for Computer Science) project, working on interdisciplinary initiatives at the intersection of AI, data science, and distributed systems.
- **Technical Skills:** Python, Research Methodology, Cybersecurity, Network Security, Distributed Systems, Docker, Git, Linux

### Pettoo UG

*Freelance Software Engineer*

Hamburg, Germany

*June 2025 - September 2025, Part Time*

- Developed and maintained the backend of a pet services platform enabling users to discover and book pet services, purchase products, and connect with other pet owners.
- Designed and managed the database architecture and implemented core business logic using FastAPI and PostgreSQL.
- Led a team of 3 developers, coordinating development tasks and ensuring consistent code quality across the project.
- Set up CI/CD pipelines using GitHub Actions and deployed services on Google Cloud Platform using Docker.
- **Technical Skills:** FastAPI, Google Cloud Platform, Supabase, Docker, Git, Github Actions, CI/CD, Elasticsearch

### University of Hamburg - Language Technology Research Group

*Project Student Assistant*

Hamburg, Germany

*Feb 2025 - May 2025, Part Time*

- Contributed to the DATS Project (Discourse Analysis Tool Suite), an NLP platform for automated text analysis and discourse processing, implementing key features and functionality enhancements.
- Developed full-stack components including a React TypeScript frontend and FastAPI backend services, with RESTful APIs and database integrations.
- Implemented NLP processing modules, text preprocessing pipelines, and data analysis algorithms to handle large-scale linguistic datasets.
- **Technical Skills:** NLP, Machine Learning, React TypeScript, FastAPI, Python, RESTful APIs, Database Integration, Docker, Git, Text Analytics, Data Processing

## PROJECTS

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### **Few-Shot Bioacoustic Classification** 🧠 | Python, PyTorch Lightning, Prototypical Networks, Hydra, MLflow

- Implemented prototypical networks for animal vocalization classification that adapts to new species from just a few labeled examples using episodic N-way, K-shot training paradigm
- Built end-to-end audio processing pipeline with offline feature extraction (STFT → Mel filterbank → Log/PCEN spectrograms) and online episodic training using PyTorch Lightning with configurable Hydra parameters and MLflow experiment tracking

### **Federated Fine-Tuning of Nicheformer** 🧠 | Python, PyTorch, Federated Learning

- Implemented federated learning framework for fine-tuning Nicheformer transformer model on mouse brain spatial transcriptomics data, comparing centralized (94.45%), federated (92.76%), and local training strategies
- Built data pipeline for anatomical non-IID partitioning (dorsal/mid/ventral regions) with 24-class cell type classification on 250-dimensional features (248 genes + spatial coordinates) achieving competitive 92.76% accuracy in federated setting

### **GitGood** 🧠 | Python, RAG, GitHub API, OpenAI API

- Built a comprehensive web application that integrates with GitHub via OAuth to analyze and visualize repository data, commit history, and code differences.
- Integrated AI-powered chat assistant using custom RAG system to answer contextual questions about code changes and repository evolution.

### **ACL Anthology RAG** 🧠 | Python, RAG, FastAPI, React, Qdrant, Langchain

- Built an intelligent research assistant that retrieves papers from 40,000+ ACL papers using dense vector embeddings and LLM-based query reformulation, enabling semantic understanding beyond keyword matching
- Designed and implemented end-to-end system with FastAPI backend (Python), React frontend, Qdrant vector database, and LangChain LLM integration with real-time streaming UI and inline citations

### **MyTorch** 🧠 | Python, NumPy, Deep Learning, Neural Networks

- Developed a minimal deep learning library inspired by PyTorch, implementing fundamental neural network operations using only NumPy as the Python library dependency, building everything from scratch with fully connected linear layers for educational and prototyping purposes.
- Built core functionalities including forward and backward propagation, gradient computation, and optimization algorithms to facilitate understanding of deep learning fundamentals.

### **Momentum** | Web Scraping, Content Analysis, Research Tools, Machine Learning, Python

- Developing a research assistant web application inspired by Anara Labs' Anara, designed to streamline academic and technical research workflows.
- Building automated web content extraction and organization systems to gather, process, and categorize research materials from multiple online sources.
- Implementing intelligent summarization algorithms to condense and present research findings in structured, actionable formats for enhanced productivity.

## TECHNICAL SKILLS

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**Languages :**Python, TypeScript, C++, SQL

**Relevant courses :**Machine Learning, Data Analytics, Natural Language Processing, Data Structures & Algorithms, Databases and Information Systems, Statistical Signal Processing, Language Technology, Computer Vision, Neural Networks

**Technologies :**PyTorch, PyTorch Lightning, Hydra, Scikit-Learn, Pandas, PostgreSQL, Langchain, Docker, Tableau, Seaborn, FastAPI, Livekit, Supabase, Redis, React, NodeJs, Github Actions, Google Cloud, Git, Tailwind CSS, MS Office

## LANGUAGES

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**English:** C1 (Advanced) | **Hindi:** Native | **Urdu:** Native | **German:** A2 (Elementary)

## AWARDS & ACHIEVEMENTS

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**Letter of Excellence and Acknowledgment**, Received from the Vice Chancellor of Jamia Millia Islamia.

**IELTS** 8.0 Band CEFR C1. 9-Reading, 8-Listening, 7.5-Writing, 7.0-Speaking

**Achieved 1st Rank** in the university for the 4th year of Bachelors (2024).